Spotify 2023 Data Analysis

Insights for music-industry decision-making using Microsoft Power BI

Songs analysed (after cleaning)	948
Total streams	487.6 billion
Median streams per song	289 million
Unique artists (incl. featured)	698
Dataset	spotify-2023.csv (953 raw rows, 24 columns)

Contents: 1 Methodology, cleaning and assumptions | 2 Music (audio) analysis | 3 Cross-platform presence | 4 Playlist exposure vs streams | 5 Chart position vs streams | 6 Outliers and exceptional songs | 7 Artist concentration (80/20) | 8 Recommendations and limitations | Appendix: Power BI report design

1. Methodology, Data Cleaning and Assumptions

The dataset was imported into Power BI (Power Query) and prepared before any modelling. The same steps were applied in Python to compute the numbers quoted in this report, so that every figure can be reproduced in Power BI using the DAX measures supplied in the accompanying build guide. Non-parametric (Spearman rank) correlation is used throughout because streams, playlist counts and chart counts are heavily right-skewed; Pearson would be dominated by a few very large songs.

Data-quality issues found and how they were handled

Issue	Evidence	Treatment
File not UTF-8	CSV failed to load with UTF-8 (Latin-1 encoding)	Imported with encoding 1252 (Latin-1)
Corrupted "streams" value	1 row (Love Grows (Where My Rosemary Goes)) had audio-feature text pasted into the streams field	Row removed (streams cannot be recovered)
Numbers stored as text	in_deezer_playlists and in_shazam_charts contained thousands commas (e.g. "2,445")	Commas removed, columns converted to whole numbers
Missing Shazam values	50 blanks in in_shazam_charts	Set to 0 (= not on Shazam chart); a flag column keeps the original missing status
Missing key	95 blanks in key	Replaced with "Unknown" (kept, key is not central to the tasks)
Duplicate songs	4 track+artist pairs appear twice (SNAP, About Damn Time, Take My Breath, SPIT IN MY FACE!) with different stats	Kept the record with the higher streams (assumed to be the more complete count)
Broken characters in names	Some track/artist names show replacement characters (e.g. Spanish accents) already damaged in the source file	Kept as is; cannot be repaired reliably. Does not affect numeric analysis
Suspicious value	"Que Vuelvas" has only 2,762 streams despite 805 playlists	Kept but flagged as a probable data error; it is the most extreme under-performer in Section 6
New fields	-	release_date, total_playlists, chart_sum, platforms_charted, cross-platform score, artist-level table

Key assumptions

• Chart columns (in_spotify_charts etc.) are treated as a **chart-presence score** where 0 = not charted and higher = stronger/longer presence. This is supported by the data: all correlations with streams are positive (if the values were ranks, where 1 is best, they would be negative). The dataset does not document the exact meaning, so conclusions on charts are stated with this caveat.

• The three playlist columns (Spotify, Apple, Deezer) are counts of playlists containing the song. Shazam has chart information only, no playlists.

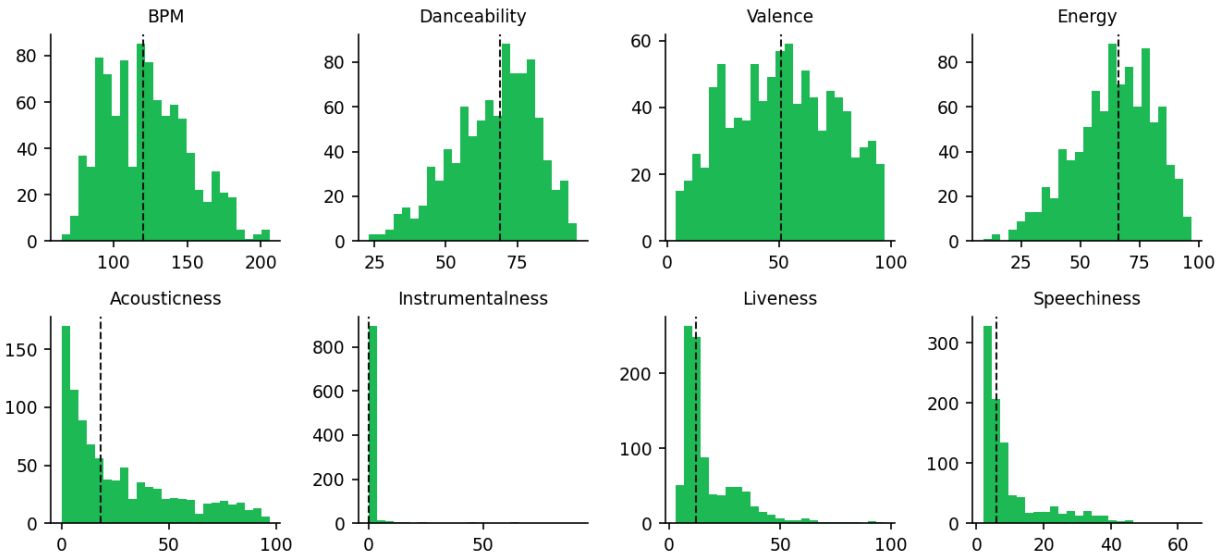
• For the artist analysis, a song with several artists is credited in full to each artist. This is standard for exposure analysis but means artist totals overlap and add up to more than the real total; percentages are calculated against the sum of artist totals.

• Streams are Spotify streams only. Playlist and chart data are a snapshot, so we cannot say whether playlists came before or after streams.

• Forecasting and predictive modelling are excluded as instructed.

2. Music Analysis - Audio Features

Chart 2.1 Distribution of each audio feature (dashed line = median). *Why histograms:* they show the shape and spread of each variable, which averages alone hide.



What the typical popular song looks like

• Popular songs are **danceable and energetic**: median danceability 69% and energy 66%, with a median tempo of 120 BPM (moderate, mid-tempo pop range).

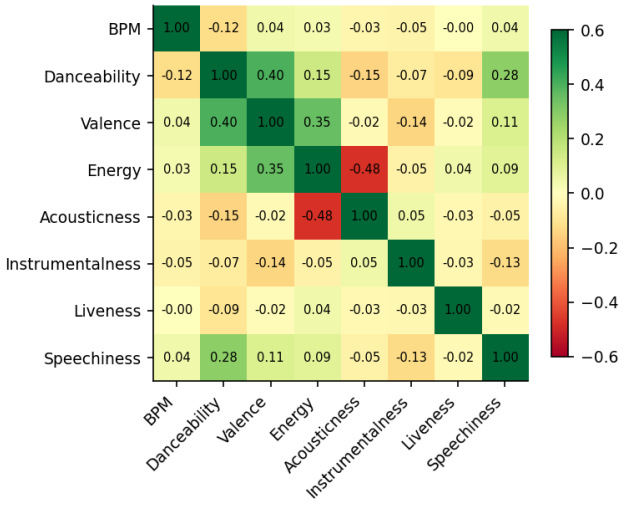
• Valence is centred near 51%, meaning moods are evenly split between happy and sad songs; there is no dominant emotional tone.

• Acousticness (median 18%), instrumentalness (median 0%), liveness (median 12%) and speechiness (median 6%) are all low and right-skewed: hit songs are mostly studio-produced, vocal-led tracks. Instrumentalness is essentially zero for almost every song.

• 58% of songs are in a major mode and 42% minor. The most common keys are C# (119 songs), G (96), G# (91) and F (89).

Relationships between features

Chart 2.2 Spearman correlation heatmap. *Why a heatmap:* it shows all pairs of eight variables in one view.



Reading the heatmap

• **Danceability and valence (+0.40)**: the strongest link. Danceable songs tend to sound more positive.
• **Energy and acousticness** are strongly negative: acoustic songs are calmer. Energy and valence are positively related.
• **Danceability and speechiness (+0.28)**: rap-style tracks tend to be rhythmic.
• **BPM** is nearly independent of all other features, so tempo does not predict mood or energy here.

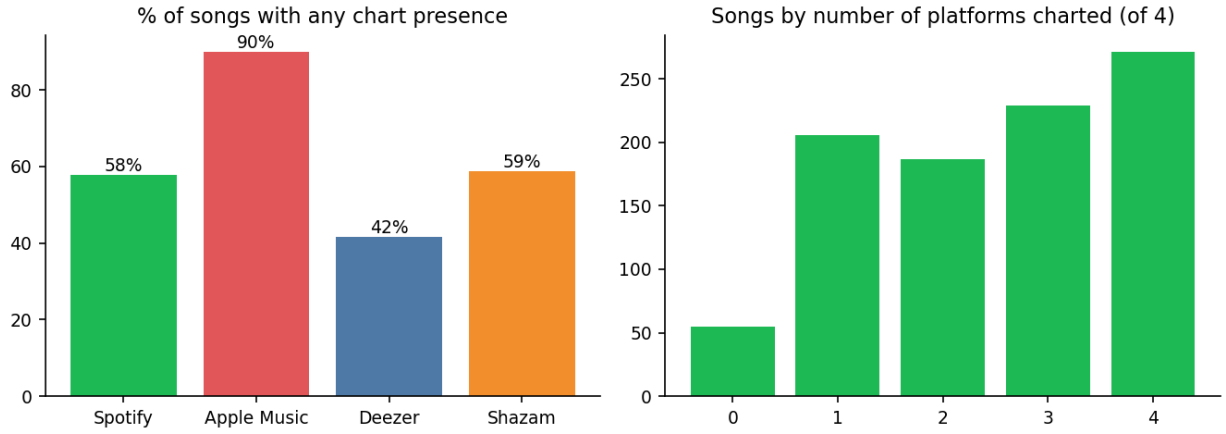
Do audio features explain popularity?

Very little. Spearman correlations between each audio feature and streams are all within ±0.11 (strongest: speechiness -0.11, danceability -0.08). Comparing the top and bottom streaming quartiles: the most-streamed quartile has slightly **lower** danceability (64.6% vs 67.5%), lower speechiness (8.1% vs 11.3%) and lower valence (48.8% vs 51.2%). These gaps are small. **Interpretation:** since every song in the dataset is already a hit, the sound of a song is fairly uniform and does not separate bigger hits from smaller ones. Exposure and artist reach (Sections 4 and 7) matter far more.

Trend over time (association only): songs released in 2023 are more danceable (70.0%) and energetic (68.2%) than pre-2020 songs (60.7% and 62.2%). This suggests that the current charting catalogue favours upbeat, danceable production, but note the pre-2020 group contains only older songs that survived to 2023, so it is not a like-for-like comparison.

3. Cross-Platform Presence

Chart 3.1 Chart presence by platform and number of platforms per song. *Why bar charts:* they compare discrete categories cleanly.



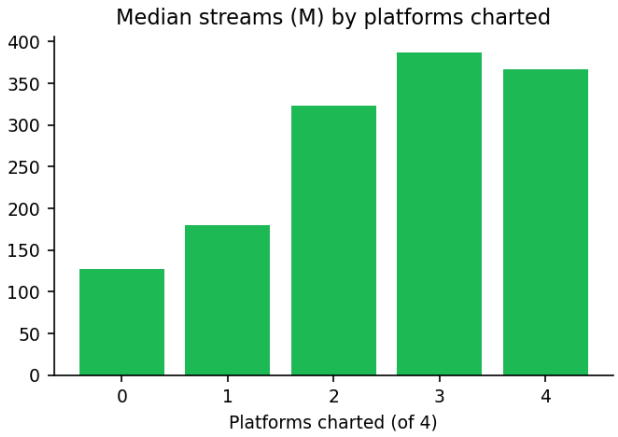
Platform differences

• **Apple Music** has the widest chart presence (89.9% of songs), followed by Shazam (58.8%), Spotify (57.7%) and Deezer (41.7%). Deezer is the most selective.

• Median playlist counts differ greatly by platform: Spotify 2,216, Deezer 44, Apple 34. These are not comparable in absolute terms because each platform has a different scale and catalogue.

• Playlist counts move together across platforms (Spotify-Deezer rho = 0.85, Spotify-Apple 0.78), while chart presence forms a separate cluster (Spotify-Deezer charts 0.59, Spotify charts-Shazam 0.54). Editorial/user playlisting and chart momentum are therefore two different kinds of exposure.

Multi-platform songs: 271 songs (29%) chart on all four platforms and only 55 on none. Median streams rise from 127.0M (no platform) to 323.0M (two) and about 387.0M (three), but flatten between three and four platforms (367.0M). Being on more platforms is associated with more streams up to a point; beyond three there is little extra gain.



Top songs by cross-platform score

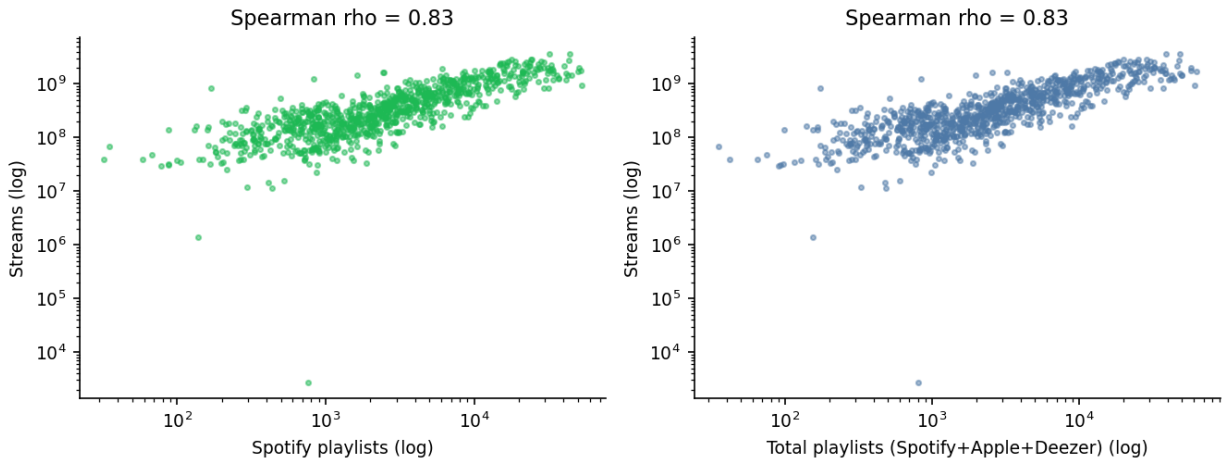
The score is the average percentile rank of a song across the 3 playlist and 4 chart measures (0-100), so no single large-scale platform dominates.

Song	Artist(s)	Streams (M)	Platforms charted	Score
Flowers	Miley Cyrus	1,317	4	96.2
Cold Heart - PNAU Remix	Dua Lipa, Elton John, Pnau	1,605	4	95.6
I'm Good (Blue)	Bebe Rexha, David Guetta	1,109	4	95.3
I Ain't Worried	OneRepublic	1,086	4	92.7
Makeba	Jain	165	4	92.6
Sunflower - Spider-Man: Into the Spider-Verse	Post Malone, Swae Lee	2,808	4	91.0
Starboy	The Weeknd, Daft Punk	2,566	4	90.9
Counting Stars	OneRepublic	2,011	4	90.9
Where Are You Now	Lost Frequencies, Calum Scott	973	4	90.0
Anti-Hero	Taylor Swift	1,000	4	90.0

Observation: "Makeba" (Jain) has a top-five cross-platform score but only 165M streams, so strong presence everywhere does not guarantee top Spotify streams. Songs such as "Sunflower" and "Starboy" combine both.

4. Playlist Exposure and Streaming Performance

Chart 4.1 Streams against playlist counts (both on log scales). *Why scatter plots on log scales:* both variables span several orders of magnitude, and a linear scale would squash 90% of songs into the corner.

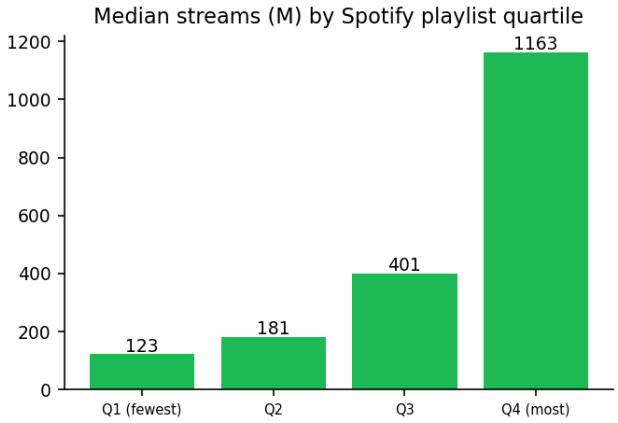


Findings supported by the data

• Playlist exposure is the **strongest predictor of streams** found in the dataset: Spotify playlists rho = 0.83, total playlists rho = 0.83, Deezer 0.74, Apple 0.67.

• A log-log regression gives R^2 = 0.62 (slope 0.68): a 10x increase in Spotify playlists is associated with roughly a 4.8x increase in streams. The slope below 1 shows diminishing returns.

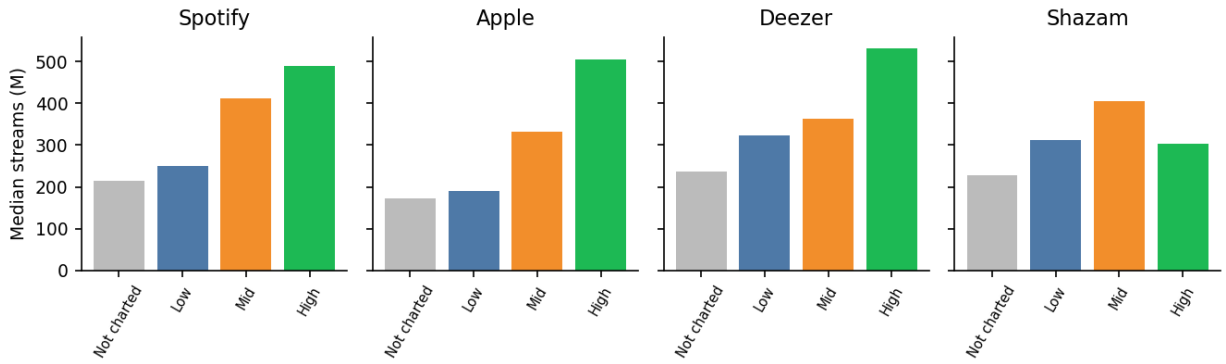
• Songs in the top Spotify-playlist quartile have median streams of 1,163M against 123M for the bottom quartile, about 9 times higher.



Association vs causation: the relationship is almost certainly two-way. Being placed on playlists exposes a song to more listeners (playlists -> streams), but popular songs are also added to more playlists because they are already streaming well (streams -> playlists). The dataset is a single snapshot, so it cannot separate these effects. The correlation shows that playlists and streams move together, not that adding playlists would produce a specific increase in streams. Unusual songs that break the pattern are examined in Section 6.

5. Chart Position and Song Performance

Chart 5.1 Median streams by chart tier on each platform. Charted songs are split into Low / Mid / High thirds of the chart score; "Not charted" = 0. *Why tiers:* chart values are heavily skewed with many zeros, so grouping makes the pattern clear.



Platform	Spearman rho (all songs)	Spearman rho (charted songs only)	Median streams: not charted	Median streams: High tier
Spotify	0.26	0.17	214M	490M
Apple Music	0.34	0.35	172M	506M
Deezer	0.23	0.16	237M	532M
Shazam	0.12	0.02	228M	303M

Interpretation

• Chart performance is **positively but only moderately** related to streams (rho 0.12-0.34), much weaker than playlist exposure (0.83). Chart presence explains a modest share of streaming differences.

• **Apple Music** is the most informative chart: the correlation stays at 0.35 even among charted songs, and median streams rise from 172M (not charted) to 506M (high tier).

• **Spotify and Deezer** show a clear step between 'not charted' and charted, and a smaller increase inside the charted group (rho 0.17 and 0.16): being on the chart at all matters more than how strongly.

• **Shazam** behaves differently: the correlation among charted songs is 0.02, and the High tier (303M) has lower median streams than the Mid tier (405M). Shazam measures song discovery (people identifying a track), which often leads streaming rather than following it, so a strong Shazam value can mean 'new and trending' rather than 'already big'. This is a possible interpretation, not something the dataset proves.

Top 10 songs by streams and their chart scores

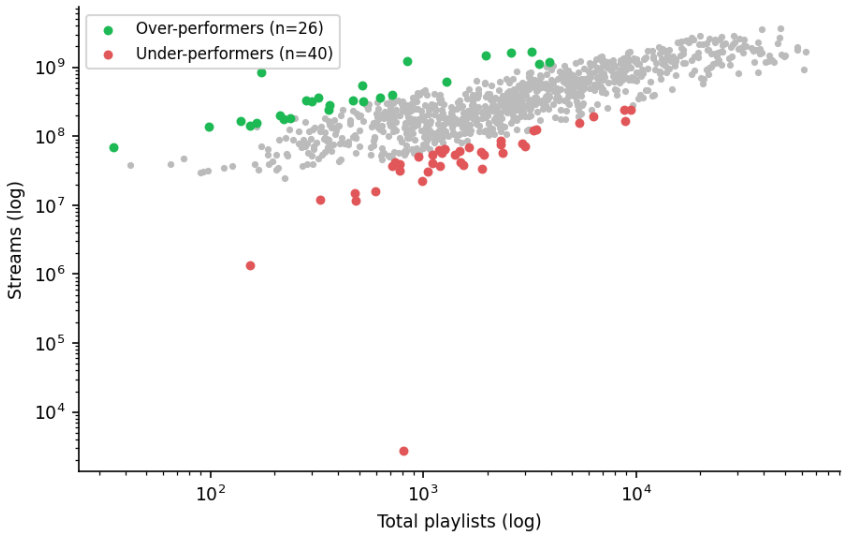
Rank	Song	Streams (M)	Spotify ch.	Apple ch.	Deezer ch.	Shazam ch.
1	Blinding Lights	3,704	69	199	20	0.0
2	Shape of You	3,563	10	0	7	0.0
3	Someone You Loved	2,887	53	125	0	0.0
4	Dance Monkey	2,865	0	167	6	0.0
5	Sunflower - Spider-Man: Into the Spider-Verse	2,808	78	117	4	69.0
6	One Dance	2,714	24	107	0	26.0
7	STAY (with Justin Bieber)	2,665	36	99	31	0.0
8	Believer	2,594	23	121	10	31.0
9	Closer	2,591	0	159	0	44.0
10	Starboy	2,566	79	137	1	140.0

The most-streamed songs of all time on the platform do not necessarily have the highest 2023 chart scores, since chart presence in this dataset reflects current momentum while streams are a cumulative total. Older catalogue

hits can lead on streams with modest chart scores.

6. Outlier and Exceptional Song Analysis

Method: a log-log regression predicts expected streams from total playlists (Spotify + Apple + Deezer). Songs whose actual streams are more than 1.5 standard deviations above the prediction are **over-performers** (high streams for low exposure); more than 1.5 SD below are **under-performers** (strong exposure but low streams). *Why residuals:* they measure how surprising a song is relative to the general pattern.



The analysis flags **26 over-performers** and **40 under-performers** out of 948 songs.

Metric (median)	Over-performers	All songs	Under-performers
Streams (M)	329	289	54
Total playlists	360	2,306	1,334
Chart score sum	39.0	66.5	35.0
Danceability %	60.0	69.0	66.5
Energy %	51.5	66.0	68.0
Acousticness %	38.5	18.0	29.0
Valence %	43.5	51.0	58.5
Release year	2022	2022	2022

Examples

Group	Song	Artist(s)	Streams (M)	Playlists	Std. residual
Over	Dream On	Aerosmith	839	173	4.0
Over	Enemy - from the series Arcane League of Legends	Imagine Dragons, League of Legends, Arcane	1,223	840	3.0
Over	Locked Out Of Heaven	Bruno Mars	1,481	1978	2.5
Over	Die For You	The Weeknd	1,648	2575	2.4
Over	Es un Secreto	Plan B	541	515	2.3
Under	Que Vuelvas	Carin Leon, Grupo Frontera	0	805	-15.4
Under	Jhoome Jo Pathaan	Arijit Singh, Vishal Dadlani, Sukriti Kakar, Vis	1	153	-5.0
Under	QUEMA	Sog, Ryan Castro, Peso Pluma	12	480	-3.1
Under	Mother I Sober (feat. Beth Gibbons of Portishead	Kendrick Lamar, Beth Gibbons	33	1895	-2.9
Under	S91	Karol G	16	598	-2.8

Supported by the data

• Over-performers have **far fewer playlists** (median 360 vs 2,306) yet slightly higher streams than the typical song (329M vs 289M).

• Their audio profile differs: **lower energy (51.5% vs 66.0%), lower danceability (60.0% vs 69.0%) and higher acousticness (38.5% vs 18.0%)**. They are also mostly solo-artist releases (median artist count 1).

• The group includes classic catalogue songs such as Aerosmith's 'Dream On' and Bruno Mars' 'Locked Out Of Heaven', and TV/film-linked songs (Arcane's 'Enemy').

• Under-performers have plenty of playlists (median 1,334) but only 54M median streams. They are mostly recent: 22 of 40 were released in 2022 and 12 in 2023. Only 40% have Shazam presence (vs 59% overall).

Possible interpretations (not proven by the data)

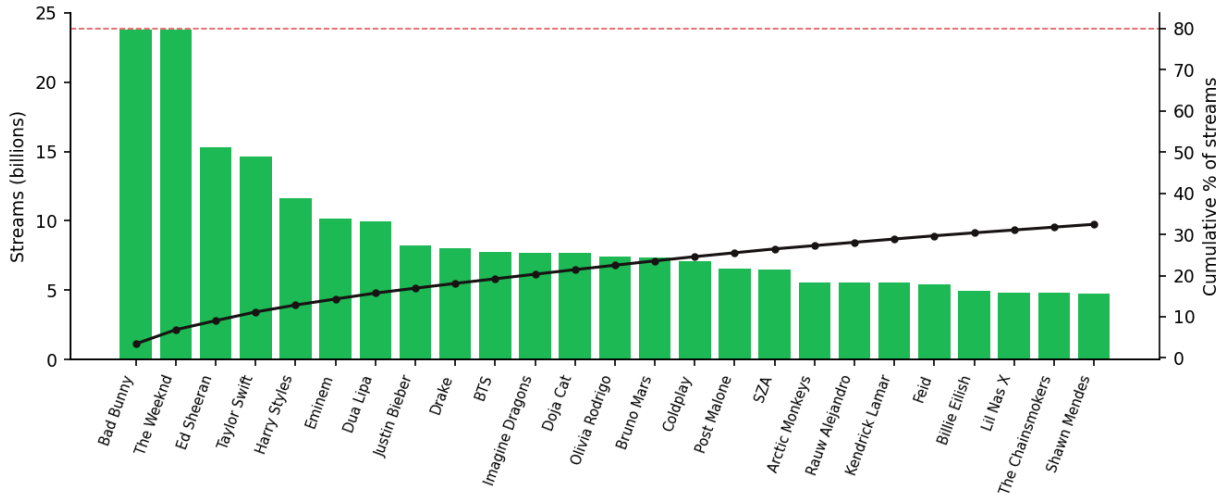
• Over-performers may be **catalogue or sync-driven songs**, whose streams accumulate from years of organic listening, radio, film/TV use and personal libraries, which do not require many playlist placements. The data supports the lower playlist counts and the older, more acoustic profile, but not the reasons for the listening.

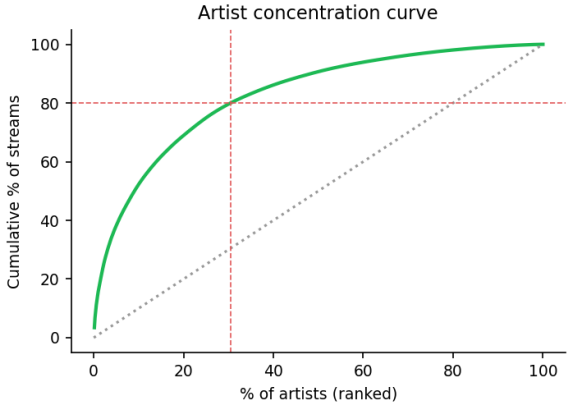
• Under-performers may be **newly released songs** that were quickly placed on many playlists (for example by a label push or by tracking a big artist) before the streams had time to accumulate. This fits their recent release dates, but release dates alone cannot confirm it.

• Some under-performers may reflect **data errors**: "Que Vuelvas" shows 2,762 streams against 805 playlists (residual -15 SD), which is implausible and should be verified against the source.

7. Artist Concentration and 80/20 (Pareto) Analysis

Chart 7.1 Pareto chart of the top 25 artists: bars show streams, the line shows cumulative % of total artist streams, and the dashed line marks 80%. *Why Pareto:* it combines ranking and cumulative contribution to show concentration.





Concentration results

• **213 of 698 artists (30.5%)** generate 80% of streams.
• Top 1% of artists = **14.3%** of streams.
• Top 10% = **52.0%**; top 20% = **68.8%**.
• Gini coefficient = **0.653** (0 = equal, 1 = one artist has everything).
• 68.1% of artists have only one song in the dataset.
• At song level, 389 songs (41.0%) produce 80% of streams.

Top 10 artists

Rank	Artist	Streams (bn)	Songs	% of total
1	Bad Bunny	23.8	40	3.4%
2	The Weeknd	23.8	36	3.4%
3	Ed Sheeran	15.3	14	2.2%
4	Taylor Swift	14.6	38	2.1%
5	Harry Styles	11.6	17	1.7%
6	Eminem	10.2	9	1.5%
7	Dua Lipa	10.0	9	1.4%
8	Justin Bieber	8.2	7	1.2%
9	Drake	8.0	19	1.2%
10	BTS	7.8	13	1.1%

Interpretation and industry implications

• Streaming is **concentrated but not a strict 80/20**: it is closer to a 30/80 pattern (30.5% of artists earn 80%) because the dataset only contains already successful songs. In the full Spotify catalogue, concentration would be far more extreme; this dataset understates it (selection bias).

• There is no single dominant artist: the top 3 (Bad Bunny, The Weeknd, Ed Sheeran) account for only 9.1% of streams. The success is a **long head** of superstars rather than a monopoly.

• Top artists succeed through **volume** (Bad Bunny has 40 songs, Taylor Swift 38, The Weeknd 36) or **scale per song** (Ed Sheeran, Harry Styles and Dua Lipa have fewer songs but very large streams). This gives labels two distinct routes: catalogue depth and hit potency.

• For the industry, this means catalogue and rights value are concentrated in a relatively small group of artists, giving them strong negotiating power, while the long tail of one-song artists represents high-variance, lower-volume opportunities.

• **Caution:** because collaborations are credited to every artist, totals overlap and concentration percentages are approximate.

8. Conclusions, Recommendations and Limitations

Summary of findings

• **Music:** hit songs are mid-tempo, danceable, energetic, vocal-led and studio-produced; audio features barely differ between bigger and smaller hits.

• **Platforms:** Apple Music has the widest chart reach, Deezer the narrowest; 29% of songs chart on all four platforms and streams stop rising after about three platforms.

• **Playlists:** playlist exposure is strongly related to streams (rho 0.83, R^2 0.62 on log scale) with diminishing returns.

• **Charts:** chart presence is moderately related to streams; Apple Music is the most informative and Shazam behaves as a discovery signal.

• **Outliers:** older, more acoustic catalogue songs over-perform relative to playlists; recently released, heavily playlisted songs under-perform.

• **Artists:** 30.5% of artists generate 80% of streams; the top 10% of artists generate 52.0%.

Recommendations for the music industry

• Prioritise playlist strategy for new releases, since it has the strongest association with streams, but monitor for diminishing returns after the first few thousand placements.

• Use Apple Music charts as a reliable performance indicator and Shazam as an early discovery signal rather than a measure of scale.

• Do not rely on audio characteristics alone when selecting songs to promote; artist reach and exposure explain much more.

• Track catalogue songs separately: they can reach high stream totals with little playlist support.

• Diversify artist portfolios: dependence on a small group of top artists concentrates commercial risk.

Limitations

• Snapshot data: no time series, so cause and effect cannot be established and forecasting is out of scope.

• Only successful songs are included (selection bias); results may not generalise to all music.

• The meaning of the chart columns is inferred, and some source records were corrupted or duplicated (see Section 1).

• Streams are Spotify-only, so comparisons with other platforms use their playlist and chart measures only.

Appendix: Power BI Report Design

Page	Visuals (with justification)	Slicers / interaction
1 Overview	KPI cards (songs, total streams, median streams, artists); top artists bar; streams by release year column chart	Year, key, mode, artist count
2 Music Analysis	Histograms (distribution), correlation matrix (matrix with conditional formatting), scatter danceability vs valence, radar-style bar of averages by stream quartile	Release year, mode, stream tier
3 Cross-Platform	Clustered bars for chart presence, column chart by platforms charted, table of top cross-platform songs with data bars	Number of platforms, year
4 Playlists vs Streams	Scatter (log axes) with trendline, quartile column chart, KPI card for correlation	Playlist tier, artist count
5 Chart vs Streams	Clustered columns by chart tier per platform, ranking table (Top N), scatter chart score vs streams	Platform, tier
6 Outliers	Scatter coloured by outlier group, comparison matrix, table of over/under-performers	Outlier group, year
7 Artist 80/20	Pareto (columns + cumulative line), KPI cards for artist count to reach 80%, Gini, top-10 table	Top N artist parameter
8 Insights	Text boxes with conclusions and assumptions	-